

Customer Behavior Analysis

Logistic Regression and Support Vector Machines

Reproducible analysis: preprocessing is fit only on stratified training data; three-fold cross-validation on a stratified training sample compares models; final results are measured once on an untouched held-out test set.

Online shoppers

12,330 rows | 17 predictors | positive class 15.5% | duplicate rows 125

Best CV model: RBF SVM (positive F1 0.649 +/- 0.041).

Held-out test: F1 0.634, average precision 0.630, ROC-AUC 0.894.

Bank churn

10,000 rows | 10 predictors | positive class 20.4% | duplicate rows 0

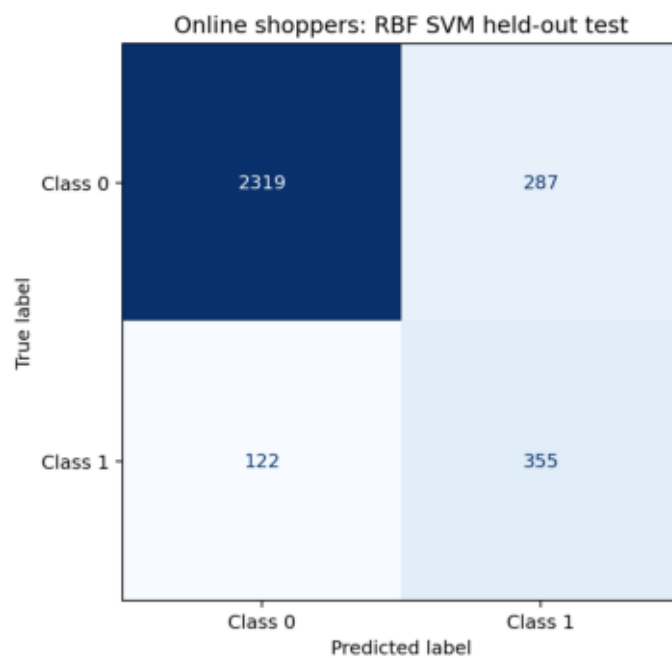
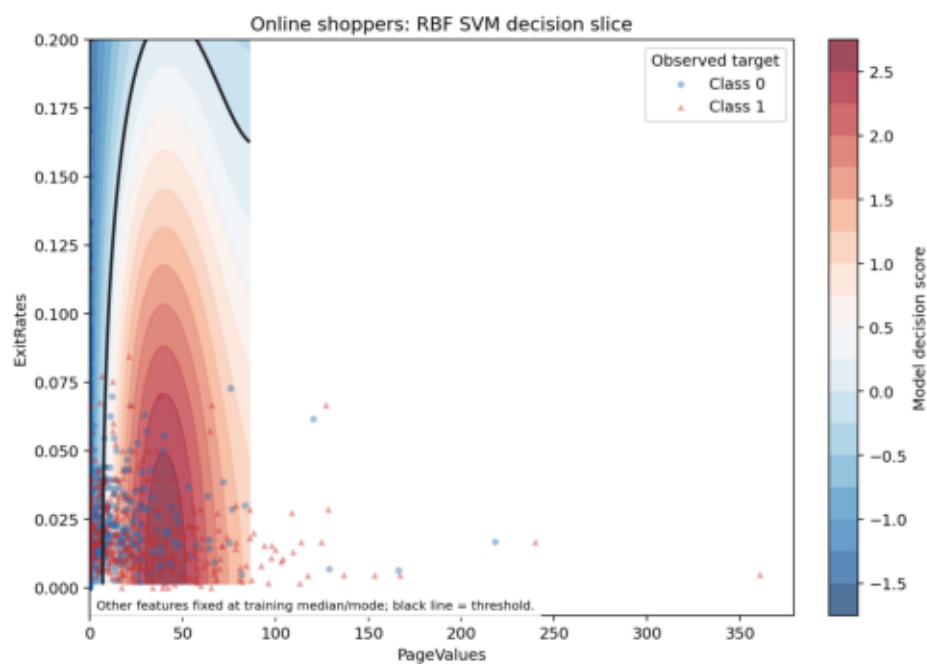
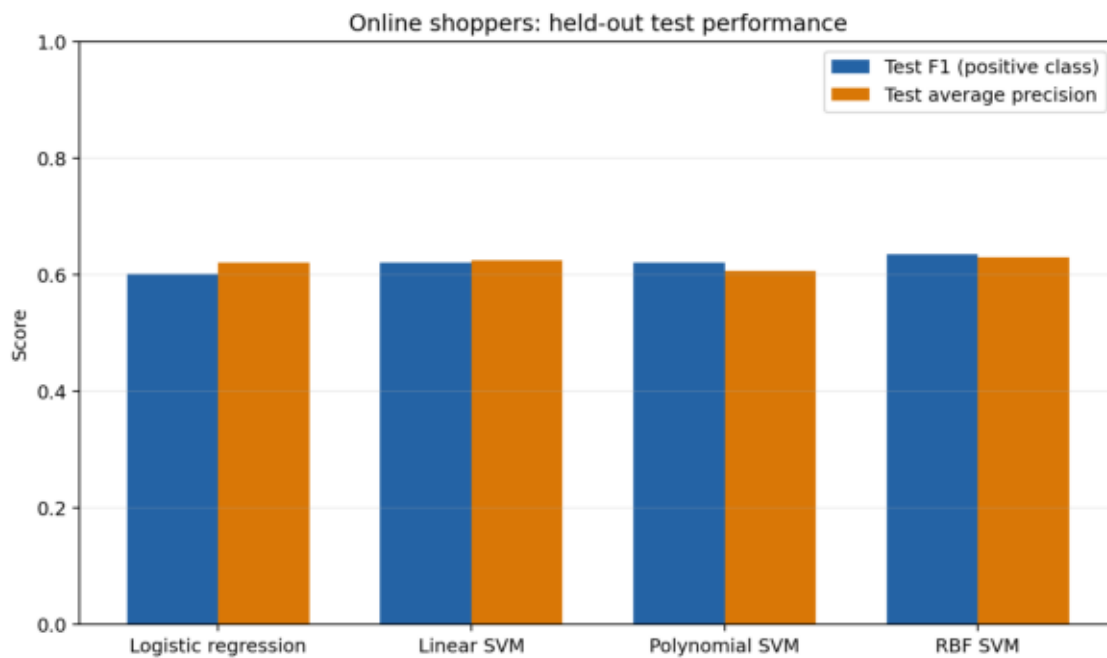
Best CV model: RBF SVM (positive F1 0.537 +/- 0.046).

Held-out test: F1 0.599, average precision 0.676, ROC-AUC 0.859.

Interpretation

Balanced class weights are used as a reproducible baseline. Charts use a valid two-dimensional decision slice: every unplotted feature is fixed to a representative training value, avoiding fabricated customer profiles.

Online shoppers: validated results



Bank churn: validated results

